

Crater geometry and ejecta thickness of the Martian impact crater Tooting

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(Received 25 October 2006; revision accepted 04 March 2007)

Abstract—We use Mars Orbiter Laser Altimeter (MOLA) topographic data and Thermal Emission Imaging System (THEMIS) visible (VIS) images to study the cavity and the ejecta blanket of a very fresh Martian impact crater ~29 km in diameter, with the provisional International Astronomical Union (IAU) name Tooting crater. This crater is very young, as demonstrated by the large depth/diameter ratio (0.065), impact melt preserved on the walls and floor, an extensive secondary crater field, and only 13 superposed impact craters (all 54 to 234 meters in diameter) on the ~8120 km² ejecta blanket. Because the pre-impact terrain was essentially flat, we can measure the volume of the crater cavity and ejecta deposits. Tooting crater has a rim height that has >500 m variation around the rim crest and a very large central peak (1052 m high and >9 km wide). Crater cavity volume (i.e., volume below the pre-impact terrain) is ~380 km³ and the volume of materials above the pre-impact terrain is ~425 km³. The ejecta thickness is often very thin (<20 m) throughout much of the ejecta blanket. There is a pronounced asymmetry in the ejecta blanket, suggestive of an oblique impact, which has resulted in up to ~100 m of additional ejecta thickness being deposited down-range compared to the up-range value at the same radial distance from the rim crest. Distal ramparts are 60 to 125 m high, comparable to the heights of ramparts measured at other multi-layered ejecta craters. Tooting crater serves as a fresh end-member for the large impact craters on Mars formed in volcanic materials, and as such may be useful for comparison to fresh craters in other target materials.

INTRODUCTION

Since the earliest days of the Viking Orbiter missions, analysis of the lobate deposits surrounding many Martian impact craters has suggested the presence of water or ice within the top kilometer or so of the crust at the time of crater formation (Carr et al. 1977; Gault and Greeley 1978; Mouginis-Mark 1979). The terms “rampart crater” and “multi-layered ejecta craters” have been used to describe these craters on Mars (Barlow et al. 2000). Impact craters on any planetary surface provide crucial information on the subsurface structure of the target material, as well as insights into the cratering process. In the case of Martian craters, the three-dimensional structure is particularly important as it may provide evidence for spatial or temporal variations in the distribution of subsurface volatiles. But due to the presence of these volatiles and an atmosphere (both of which increase the rate of modification of the crater), one particularly challenging task is to determine the original morphological attributes of fresh craters prior to their modification by erosion and/or deposition.

Using images from the Thermal Emission Imaging System (THEMIS) and the Mars Orbiter Camera (MOC), we

have identified a 29 km diameter crater located at 23.4°N, 207.5°E that meets many of the criteria for having formed in the very recent past, perhaps within the last few million years (Mouginis-Mark et al. 2003). This crater, which has provisionally been given the name Tooting crater by the International Astronomical Union (after a town in southwestern London, England), lies to the west of the Olympus Mons aureole and formed in lava flows from the volcano (Fig. 1a). Tooting crater has been the target of numerous imaging opportunities by the THEMIS Science Team, providing almost complete coverage at a spatial resolution of 18 m/pixel (Fig. 2). Tooting crater is a multi-layered ejecta crater, based upon the classification of Barlow et al. (2000). Inspection of the 30 THEMIS VIS images of Tooting that have been collected by the end of 2006 (Fig. 2) reveals that there is a total of 13 superposed impact craters on the ejecta blanket, which has an area of ~8120 km². These craters have diameters ranging between 54 and 234 meters (equivalent to 3 to 13 THEMIS pixels). We binned the number of craters based upon their diameter, and for impacts superposed upon the Tooting crater ejecta blanket in the diameter range 88 to 125 meters, the crater density is ~0.00049 1/km² and for the craters in the diameter range 125

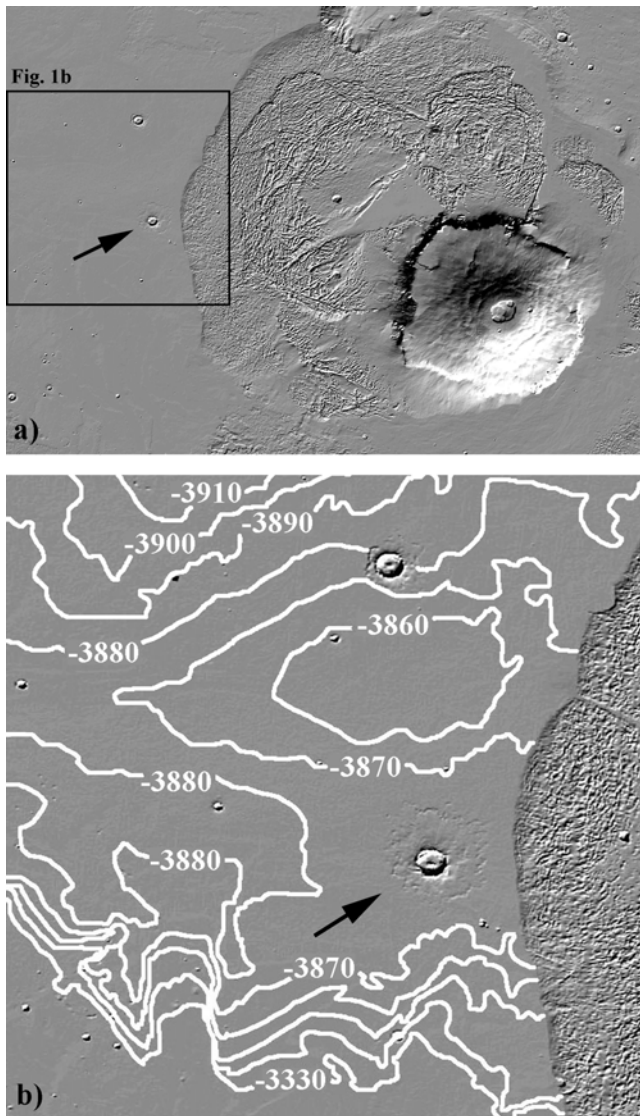


Fig. 1. a) Location map for Tooting crater (arrowed), which lies just to the west of the Olympus Mons aureole at 23.4°N, 207.5°E. Base image is a MOLA shaded relief version of the topographic data set from 10°–35°N, 200°–235°E. The black rectangle outlines the area shown in (b). b) Topography of the area surrounding Tooting crater (arrowed), derived from the MOLA 128th degree DEM. Contours are at 10-meter intervals and provide elevations relative to the Mars datum. Note that the contours for the Olympus Mons aureole materials have been omitted for clarity. Base image is a MOLA shaded relief image.

to 176 m the crater density is ~ 0.00074 . Using the final 2004 iteration of the Martian crater-count isochron (see Table 2 in Hartmann 2005), this gives an approximate age for Tooting crater that is most likely less than 2 Myr, and is probably in the range of 0.4 to 1.7 Myr. Our analysis of this morphologically fresh crater not only provides information on the original geometry of large impact craters on Mars, but also enables the original morphology of the crater cavity and ejecta layers to be examined. In addition, it places constraints

on the recent subsurface distribution of volatiles, and because of the relatively young volcanic target rocks (Scott and Tanaka 1986; Fuller and Head 2002) has potential implications for the source area of at least some of the basaltic SNC meteorites and the petrologic interpretation of these rocks (Mouginis-Mark et al. 1992; McSween et al. 1996; Kring et al. 2003).

Tooting crater formed on virtually flat lava flows within Amazonis Planitia (Fig. 1b) where there appear to have been no other major topographic features prior to the impact. The crater formed in a flat area $\sim 185 \times 135$ km in size that is at an elevation between -3870 m and -3880 m relative to the Mars datum. Locally, just beyond the ramparts of the ejecta layers in all azimuths, the elevation varies from -3870 m to -3874 m. The depth of the crater and the thickness of the ejecta blanket can therefore be determined by subtracting the appropriate elevation of the surrounding landscape (-3872 m) from the individual MOLA measurements. Thus, for the first time, it is possible to determine the absolute ejecta thicknesses for a fresh multi-layered impact crater on Mars as the radial decrease of thickness as a function of distance away from the rim crest. It is also possible to study the geometry of this fresh large crater using the MOLA measurements. As such, this study serves as a reference point against which to study the ejecta of more eroded multi-layered ejecta craters of similar size; knowledge of the radial distribution of ejecta for craters in the southern highlands may be particularly important for assessing erosion rates of craters at different periods of the history of Mars (Forsberg-Taylor et al. 2004). Tooting crater may also serve as reference point against which to compare the geometry of fresh craters of similar size that formed in other target materials such as the northern plains (Boyce and Mouginis-Mark 2006), where double-layered ejecta craters are far more common than on young lava flows such as those found in Amazonis Planitia (Barlow and Perez 2003).

CRATER CAVITY GEOMETRY

Thirty-two Mars Orbiter Laser Altimeter (MOLA) orbits cross the rim crest of Tooting crater (Fig. 3), permitting a profile around the rim crest of the crater to be determined. Figure 4 shows the location of the measured points and the variation in height around the rim crest using the individual MOLA shots. These measurements were obtained by using each individual MOLA profile that crosses the rim crest, and by identifying the highest point within each profile where it crosses the rim crest. We choose to use this shot-by-shot method for the identification of the rim height, rather than using the interpolated 128th-degree MOLA digital elevation model (DEM), because our measurements avoid using interpolated data in the DEM that might either include MOLA shots that are not the highest points on the rim, or include interpolated data between MOLA profiles. The spacing of our rim height measurements is therefore not uniform because the

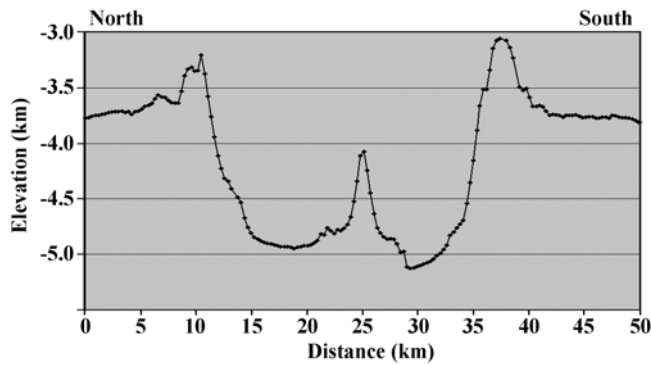


Fig. 5. Topographic profile taken through the highest point on central peak of Tooting crater. Profile derived from MOLA orbit 14,387. See Fig. 3 for location. Vertical exaggeration is $\times 10$.

highest point of the central peak is at an elevation of -4073 m relative to the datum. From other MOLA profiles, there is a secondary peak that attains an elevation of -4484 m (orbit 14,387), a broad low rise to the west of the main central peak that reaches an elevation of -4633 m (orbit 11,840), and a smaller rise to the east of the central peak reaches -4641 m (orbit 19,196). The lowest elevation of the southern crater floor is -5125 m (orbit 16,525), as measured from 24 MOLA orbits that cross this part of the crater. The lowest elevation of the northern crater floor is -4934 m (orbit 16,525), as measured from 26 MOLA orbits.

Thus, the maximum height of the central peak above the northern floor is 861 m, and the maximum height of the central peak above the southern floor is 1052 m. Central peak width is 9828 m (E-W) and 8280 m (N-S), giving an average width of 9054 m. The average height-to-base ratio of the central peak is 0.106 . The central peak width is appreciably larger than the 6065 m predicted by the relationships developed by Hale (1983) for 1672 Martian craters, and lies at the $+1$ sigma value for craters with the diameter of Tooting crater.

The measured depth below average rim crest is 1690 m for northern floor, and 1881 m for southern floor. Garvin and Frawley (1998) established the relationship $d = 0.25 D^{0.49}$, where d is the depth and D the diameter, both in kilometers, for 98 of the most unmodified Martian craters. Using this relationship, Tooting crater should have a depth of 1302 m, but in reality it is ~ 1.3 times deeper on the northern floor, and ~ 1.4 times deeper on the southern floor, than predicted by Garvin and Frawley (1998) for this diameter.

The large depth and large central peak width suggest that Tooting crater may have significantly less infill material than is typical for Martian craters of this diameter. This interpretation is also supported from analysis of MOC images of the floor (Fig. 6), which reveals that the southern floor has a polygonally fractured surface similar in morphology to the impact melt identified in lunar craters such as Tycho and Aristarchus (Strom and Fielder 1971), Copernicus (Howard

and Wilshire 1975) and King (Heather and Dunkin 2003). The northern floor appears to be more mantled, with many pits that are a few hundreds of meters in diameter on the floor (Fig. 6a), suggesting that some other type of material (the origin of which is as yet unidentified) may lie on top of an impact melt sheet.

EJECTA DISTRIBUTION

The ejecta deposits around multi-layered ejecta craters are thought to be closely tied to target properties and the ambient conditions at the time of impact. Typically, this has been attributed to the presence of volatiles (water or ice) within the target material at the time of crater formation (Carr et al. 1977; Gault and Greeley 1978; Mouginis-Mark 1979). Alternative models attributing atmospheric effects have also been proposed (e.g., Schultz and Gault 1979; Schultz 1992; Barnouin-Jha and Schultz 1998). In the case of Tooting crater, the ejecta deposit is asymmetric (Fig. 2). Presumably, this asymmetry is due to the angle of impact of the projectile that formed the crater, as there is no topographic gradient in the vicinity of Tooting crater (Fig. 1b) that could have influenced gravity-driven flow of the ejecta (Mouginis-Mark and Baloga 2006).

Few previous studies have investigated the thickness of these ejecta deposits, but the radial decay of thickness most likely is directly relevant to the emplacement process (Baloga et al. 2005; Barnouin-Jha et al. 2005; Mouginis-Mark and Baloga 2006). Once more assuming a flat pre-impact surface and a base elevation for the surrounding terrain of -3872 m relative to the MOLA data allows the spatial distribution of the thickness of the Tooting crater ejecta blanket to be explored. Point-by-point studies of the thickness of the ejecta reveals some surprising features of the Tooting crater ejecta blanket. For example, south of the crater rim (Fig. 7) the ejecta layers can be remarkably thin; there are places where the ejecta layers are between 3 to 5 m thick at radial distances of ~ 11 km from the rim crest, or $\sim 30\%$ of the maximum radial distance in that azimuth from the crater rim. Distal ramparts are 60 – 125 m high, comparable to the heights of ramparts measured at other multi-layered ejecta craters (Mouginis-Mark and Baloga 2006).

A total of $24,201$ MOLA shots from 90 individual orbits lie on the ejecta blanket beyond the rim crest of Tooting crater out to a radial distance of 66.5 km (Fig. 8). The gridded 128^{th} -degree MOLA elevation data show that the crater cavity volume (i.e., the volume of the cavity which lies below the -3872 m datum) is ~ 380 km³ and the volume of materials above the elevation of the -3872 datum is ~ 425 km³. Assuming no losses in the ejecta volume due to erosion, this implies that the amount of structural uplift of pre-existing strata (which is most important within ~ 0.8 to 1.2 crater radii from the rim crest; see p. 87 of Melosh 1989), and the bulking of the material that now forms the ejecta blanket,